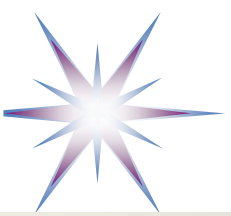


LLM/GenAI as Researcher Tool: Knowledge Retrieval, Structuring, Projection, Epistemic Validation

Knowledge Complexity model as a basis for
Human – AI Agent and Inter-Agents interaction

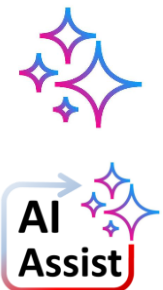
RFC - Request for Comments - Brief introduction to initiate discussion

Yuri Demchenko
IvI UvA, NTUU “Igor Sikorsky KPI”



Outline and Concepts – Planned structure

- Introduction: Background, origination and grounding of this research
 - Research, Education and Researcher potential development
- Knowledge complexity model and GenAI assisted research framework
 - ➡ • **Knowledge Complexity** and Epistemic Governance Model of LLM-Assisted Research (**KCEG**)
 - ➡ • Human-Governed *LLM Epistemic Projection* for Frontier Research (**HGLEP – HGLEP-FR**)
- KCEG applied to and tested for practical research tasks
 - Structured Systematic Literature Study (SSLR) – Knowledge complexity as metrics for paper assessment
 - Software Complexity Model and Engineering Guidance (SCEG) and Agentic Development Environment
 - Further/ongoing research: temporal aspect/dimension of KCEG, required knowledge body for edu/train
- Tasks and goals of this research
 - How to make KCEG & HGLEP common GenAI tools and methodology for Research as a literacy competence scope/base in the reality of emerging GenAI/LLM?



Sign indicating Generated with GenAI/ChatGPT
or Assisted with ChatGPT (AGI)

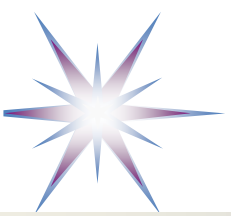
Disclaimer

This research and development is based on and built around the concept of Knowledge.

- Knowledge doesn't have the measurable properties similar to (research) experiments.
- Knowledge has variety of presentation and expression forms
- But knowledge can be validated with the epistemic correctness – attributed only to humans/researchers and dependent on their personal epistemic model

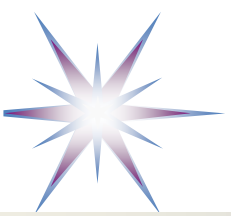
Consequently, presented here information (working knowledge) may have variable form, however clearly representing the author's epistemic model of the research topic





Setting up a playground

- Conceptual chain for Human - AI interaction based on HGLEP/KCEG
- Background Research and Development Streams: Paradigm change
- Human – AI interface
- Shared space and repositories
 - <https://github.com/COGEN-RE/Main/wiki/COGEN%E2%80%90Wiki> – Working materials and information
 - <https://github.com/COGEN-RE/About> - For general and up-to-date information
 - Free use under CC BY 4.0 License – Contributions welcome



Conceptual chain for Human - AI interaction for HGLEP/KCEG

1. Knowledge source

- paper, dataset, prior dialogue, experiment, model, human expertise

2. Knowledge presentation

- text, diagram, table, formula, code, LLM response

3. Knowledge representation

- concepts, relations, ontology, claims, evidence, assumptions, methods

4. Knowledge perception / reconstruction

- human or AI agent interprets and reconstructs the meaning

5. Knowledge assessment

- KCEG evaluates level, completeness, uncertainty, reliability, novelty

6. Internalisation / agent memory update

- human updates personal epistemic model
- AI agent updates context, memory, graph, or retrieval state

7. Knowledge action

- explanation, critique, decision, experiment, paper writing, new prompt

8. Regeneration

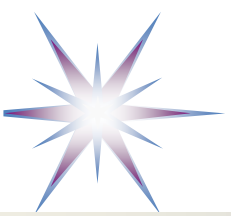
- human or AI generates a new presentation

Knowledge is an actional element of the human cognitive composition

- Knowledge structure/direct actions
- Built-in/constructed/re-constructed knowledge structure defines learning efficiency and effective curricula

Basis for a generative loop:

presentation → perception → representation → assessment → memory → action → new presentation



Human - AI Epistemic Interaction Model

Human agent:

Personal epistemic model

- semantic memory
- methodological knowledge
- tacit expertise
- epistemic judgment
- goals / intentions

AI agent:

Agent knowledge state

- context window/prompt
- retrieved knowledge
- latent model representation (ML internal)
- symbolic/graph memory
- tool outputs
- generated presentations

Interaction process:

Human intention



Prompt / instruction presentation



AI internal representation



Generated knowledge presentation



Human perception and reconstruction/representation



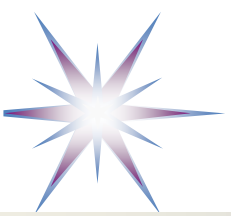
KCEG assessment (by human/researcher)



HGLEP-guided refinement

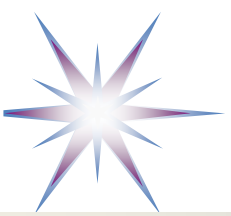


Updated human and AI knowledge states



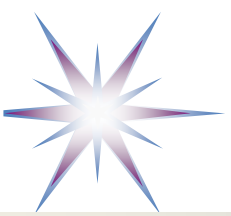
Background Research and Development Streams

- Notes: Stories and Cases
- Paradigm change: AI/GenAI/Agentic AI – By spotted comments by industry leaders
- Paradigm(s) to change:
 - GenAI/Agentic LLM is not a blackbox &
 - GenAI/LLM construct knowledge, not just retrieves stored information or data (Model is structured data used to construct knowledge) &
 - AI agent builds your cognitive model (explicitly or implicitly based on prompt)
- Preamble: Before diving into Knowledge Complexity – *What is knowledge complexity?*
- Linking Knowledge Complexity Epistemic Governance/Projection Model to Intellectual/Cognitive activity/behavior of humans:
 - **Down projection effect:** from PhD/PhD researcher down to university graduates down to school educated down to children



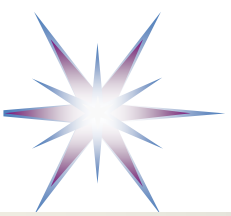
Knowledge Complexity Aspects

- General, linked to knowledge presentation
 - Vocabulary and semantics describe **how knowledge is expressed** (in words and terms).
 - Ontology and structure describe **how it is organised**.
 - Inference and methodology describe **how it works and how it is produced**.
 - **Epistemic measures describe how reliable, novel, uncertain, or contested it is.**
- For humans, complexity is strongly tied to prior knowledge, cognitive load, and research judgment.
- For LLMs, complexity is tied to representation, grounding, context, retrieval, reasoning depth, and verifiability.
- The measure cannot be a single score but a **complexity vector** across these dimensions.



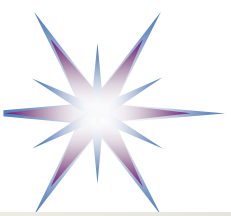
LLM – Researcher interaction – HCI -> HAI Interface

- Classical HCI provides access to resources and processes
- Human AI (HAI) Interface will provide access and interaction to the generated content/knowledge outcome to a range of work and decisions made by AGI
 - With the purpose of building and applying a user/agent cognitive model
- Expected benefit
 - Automation of many tasks
 - Optimisation and enrichment in LLM presenting knowledge to humans (operator, researcher, organisational role)
 - Team interaction and building shared knowledge corpus
 - Potentially growing a built-in reasoning/inference model as a result of interaction
 - Is the LLM Model affected or modified by HAI provenance (logging conversation)?
-



Building the Framework for LLM assisted Research

- Knowledge presentation
 - Knowledge representation
 - Knowledge perception
- Knowledge complexity *down-projection* in Perception – Representation (- Internalisation)
- Knowledge complexity assessment in application to research and educational artefacts



Knowledge Presentation – Representation - Perception

Knowledge Presentation

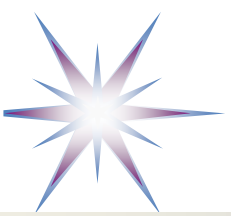
1. Vocabulary/Terminology complexity (KC1)
2. Semantic complexity (KC2)
3. Ontological complexity (KC3)
4. Structural complexity (KC4)
5. Methodological complexity (KC5)
6. Epistemic complexity/validity (KC6)

Knowledge Representation

1. Vocabulary / terminology (K1)
2. Syntax / expression form (K2)
3. Factual / propositional layer (K3)
4. Semantic layer (K4)
5. Ontological layer (K5)
6. Structural / graph layer (K6)
7. Inferential layer (K7)
8. Methodological layer (K8)
9. Epistemic / research-frontier layer (K9)

Knowledge perception

- If a human doesn't have a full Kx or KCx model of KCEG, they may treat received/observed information structured wrongly or a kind of imaginary mixed
- Children are imaginary treating objects and animals with human features
 - Agent-centered perception model by children
 - Can we control and speed up development?
- Can we facilitate cognitive development of the Doctoral (PhD) Students?
 - Hypothetically because of KCEG
- TODO: Adding Pragmatics to Semantics Complexity level as related to language understanding by humans



Knowledge Structure in LLM (to revise):

An LLM does not “know” in the same way a researcher knows

A. Parametric knowledge

What is statistically internalised in model weights during training:

- distributed
- implicit
- non-transparent
- hard to audit directly

B. Retrieved knowledge

What the model accesses from papers, databases, or knowledge bases during inference (answering):

- external
- traceable
- updateable
- more auditable

Can be enriched by using RAG.

C. Constructed working/candidate knowledge

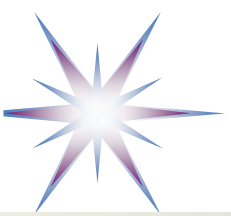
What the LLM assembles during a task:

- summaries
- concept maps
- hypotheses
- claim graphs
- argument chains

It is a **task-dependent epistemic construction** built from:

- model priors
- retrieved evidence
- instructions
- tools
- and interaction with the researcher via researcher prompt and (built) interaction context

This is not exactly pre-existing knowledge, but it can have a value as **candidate knowledge**.



KCEG Perception by Expert and Novice

Mechanism: Perception is holistic, understanding is selective reconstruction

- **1. Real object/event/system**

The full **THING EXISTS** with many properties: visible, functional, structural, causal, methodological, historical, epistemic.

- **2. Sensory/perceptual intake**

The person receives **SIGNALS**: visual form, words, diagrams, behaviour, motion, numbers, names, artefacts.

- -----

- **3. Cognitive reconstruction**

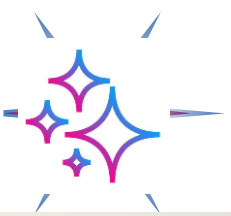
The **MIND MAPS** signals into existing *schemas*: vocabulary, concepts, relations, structures, methods, causal models, epistemic status.

- **4. Expressed understanding**

The person explains, draws, predicts, classifies, evaluates, or acts.

- **Hypothesis: Gap between layers 2 and 3/4.**

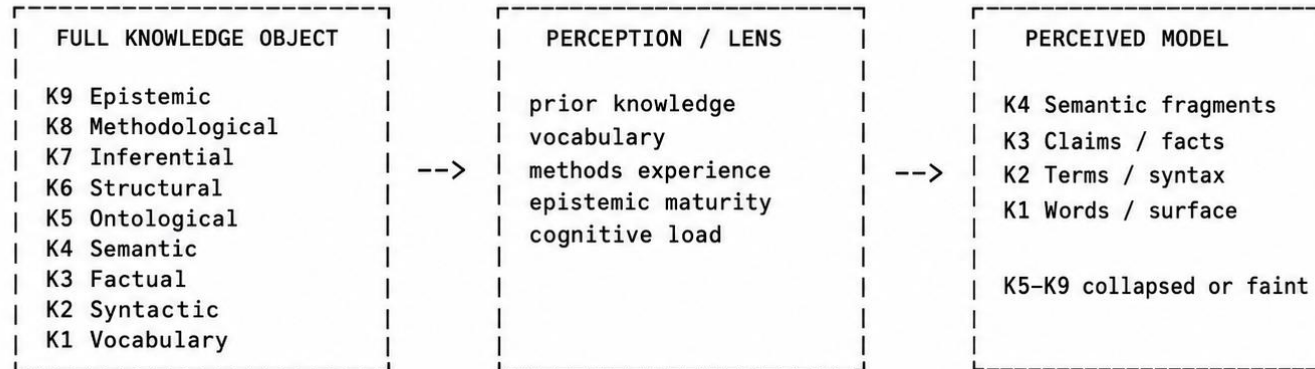
- The person may “see” the whole event, but their **model of what is significant** is constrained by their available cognitive schemas.
- This is close to the expert-novice literature review: experts detect meaningful patterns that novices overlook, and experts organise knowledge around deep principles rather than surface attributes.



KCEG Levels Perception and Down-projection (ASCII)

(1) KCEG / HGLEP Diagram Draft — ASCII / Text Sketch

Full knowledge complexity, perception lens, and down projection



K1-K9 visual progression

K1 Vocabulary	: [algorithm] [dataset] [energy]
K2 Syntactic	: $x = f(y)$, $A \rightarrow B$, if $x > y$
K3 Factual	: [Claim] $\rightarrow$ [measurement/result]
K4 Semantic	: [term] $\rightarrow$ [meaning]
K5 Ontological	: [Experiment] --uses--> [Dataset]
K6 Structural	: [Problem] $\rightarrow$ [Model] $\rightarrow$ [Evaluation]
K7 Inferential	: [Premise] $\Rightarrow$ [Inference] $\Rightarrow$ [Conclusion]
K8 Methodological	: [Question] $\rightarrow$ [Data] $\rightarrow$ [Experiment] $\rightarrow$ [Validation]
K9 Epistemic	: [Model] surrounded by {uncertainty, assumptions, novelty}

Down projection examples

K8 workflow	$\rightarrow$ perceived as "they used a method"
K7 inference	$\rightarrow$ perceived as a single conclusion
K6 structure	$\rightarrow$ perceived as section headings
K5 ontology	$\rightarrow$ perceived as isolated terms

(3) K1-K9 Legend and Notation System

Visual grammar for direct implementation in Visio / PowerPoint / draw.io

KCEG Levels Perception and Down-projection (1)

Level	Name	Node / Glyph	Relation / Notation
K1	Vocabulary	[dataset]	Loose grouping or no edge; notation: term token
K2	Syntactic	$x = f(y)$	Ordered line / sequence; notation: formal expression
K3	Factual	Claim: R increases with X	Evidence link; notation: claim -> observation
K4	Semantic	[term] $\longrightarrow$ [meaning]	Explains / means arrow; notation: interpretation
K5	Ontological	Entity A $\xrightarrow{\text{is-a}}$ Entity B $\xrightarrow{\text{part-of}}$ Entity C	Relation labels: is-a, part-of, uses, produces
K6	Structural	System $\longrightarrow$ [Module A Module B Module C]	Composition and subsystem links; notation: framework structure
K7	Inferential	Premise(s) $\xrightarrow{\text{thick}}$ Inference $\xrightarrow{\text{thick}}$ Conclusion	Thick reasoning arrows; notation: premise $\Rightarrow$ conclusion
K8	Methodological	1 $\dashrightarrow$ 2 $\dashrightarrow$ 3 $\dashrightarrow$... $\dashrightarrow$ n	Process arrows; notation: protocol / method pipeline
K9	Epistemic	? 70%	Dotted boundary / confidence band; notation: assumptions, uncertainty, novelty

Standard edge styles

1	2	3	4	5	6
plain line = association	ordered arrow = sequence	labeled arrow = ontology relation	thick arrow = inference/support	dashed arrow = process / workflow	dotted halo = uncertainty / epistemic status

Recommended color grouping

Surface / Presentation (K1-K3)	Conceptual / Representation (K4-K6)	Research / Epistemic (K7-K9)
-----------------------------------	--	---------------------------------

Knowledge Levels

1. Vocabulary / terminology (K1)
2. Syntax / expression form (K2)
3. Factual / propositional layer (K3)
4. Semantic layer (K4)
5. Ontological layer (K5)
6. Structural / graph layer (K6)
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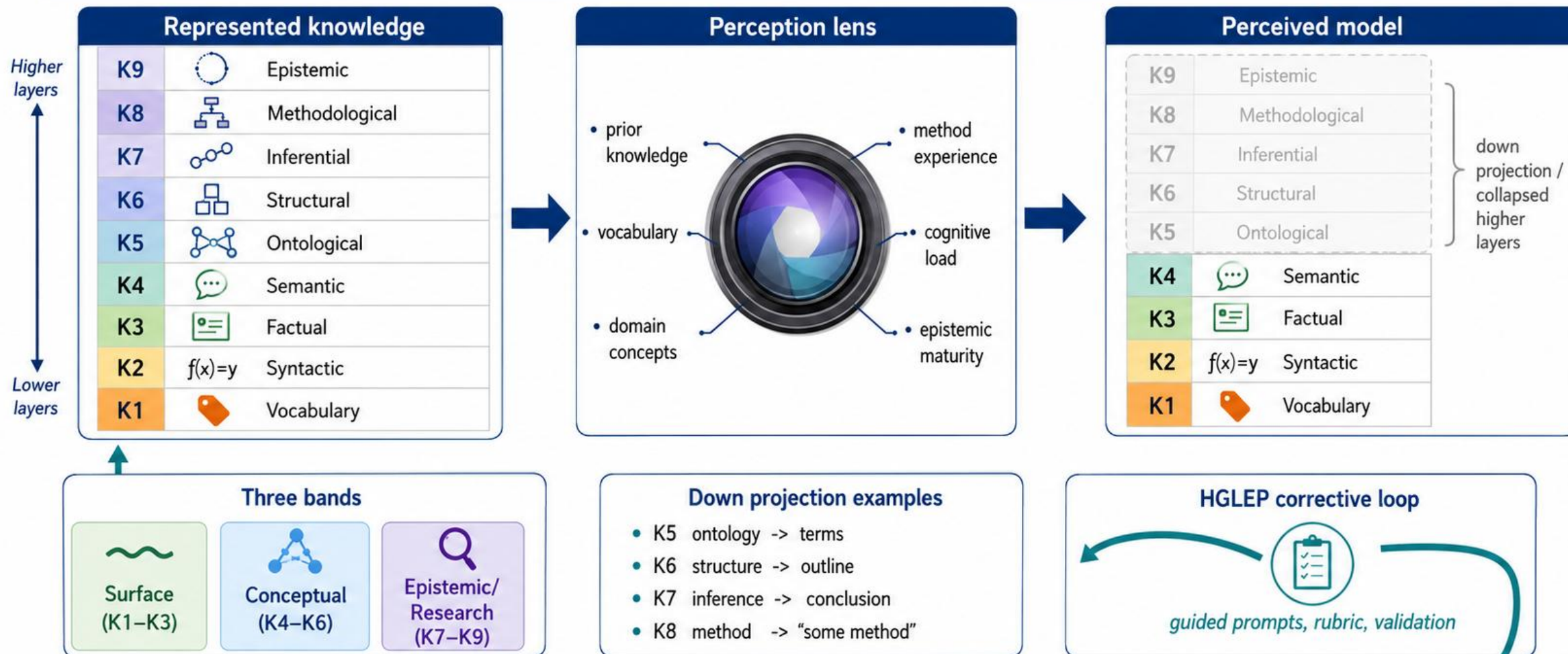


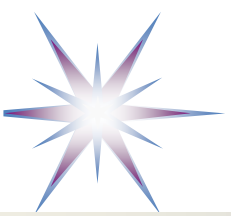
(2) One-Slide PowerPoint Layout — KCEG Projection Model

From represented knowledge to perceived understanding



A knowledge object may contain K1–K9 complexity, but a beginner may reconstruct only a K1–K4 or K1–K6 version.



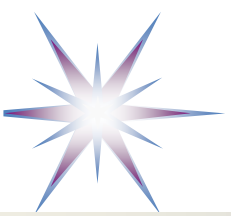


Proposed Framework: Human-Governed LLM Epistemic Projection for Frontier Research

- Human-Governed LLM Epistemic Projection for Frontier Research (HGLEP – HGLEP-FR)
- **LLM-assisted research** is most valuable when the LLM acts as an **epistemic projection instrument**: it transforms uneven, large-scale, and partially reliable accessible knowledge into structured **research candidates**, while the expert researcher governs interpretation, verification, and integration into validated knowledge.

== Based on ==

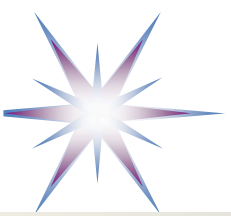
- Knowledge-Complexity and Epistemic-Governance Model of LLM-Assisted Research (KCEG)
- LLM should not be treated as autonomous producers of validated research knowledge.
 - In frontier research, LLM can **project, reorganize, and structure LLM-accessible knowledge** around a researcher's question.
 - The resulting outputs are not final truths, but **candidate knowledge structures** that require expert interpretation, methodological verification, and epistemic validation.
- The researcher remains the **epistemic governor** of the process.



LLM for Scientific Research

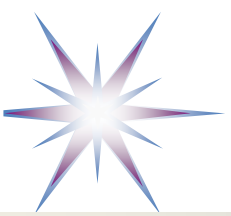
- LLMs as moving from **task-specific automation** toward **increasingly autonomous scientific agents**
 - **LLM4SR: A Survey on Large Language Models for Scientific Research** (arXiv:2501.04306v1)
 - LLM use across hypothesis discovery, experiment planning/implementation, scientific writing, and peer review
- **Level A: LLMs as productivity tools**
 - Supports summarisation, search, drafting, coding, translation, data extraction, and review assistance.
 - Valuable but not conceptually revolutionary by itself.
- **Level B: LLMs as knowledge-structure engines**
 - LLMs help transform literature into **maps of concepts, claims, evidence, assumptions, contradictions, methods, and open problems**.
 - A basis for vocabulary/semantics/ontology/structure framework
 - Provides a structured knowledge summary for the Research Question (aligned with the domain knowledge)
- **Level C: LLMs as epistemic agents**
 - This is the frontier: systems that propose hypotheses, design tests, evaluate evidence, update beliefs, and decide what experiment or analysis should happen next.
 - Requires measuring not only output quality but **scientific judgment**.
- (Revisit) Level D: Elisa LLM as psychological partner or counterpart (for high intensity cognitive load with GenAI)

Subject for research investigation: Implicit/Explicit Research cognitive profile evolution/development facilitated by LLM structured knowledge provisioning (as a result of LLM sycophancy beyond hallucination).

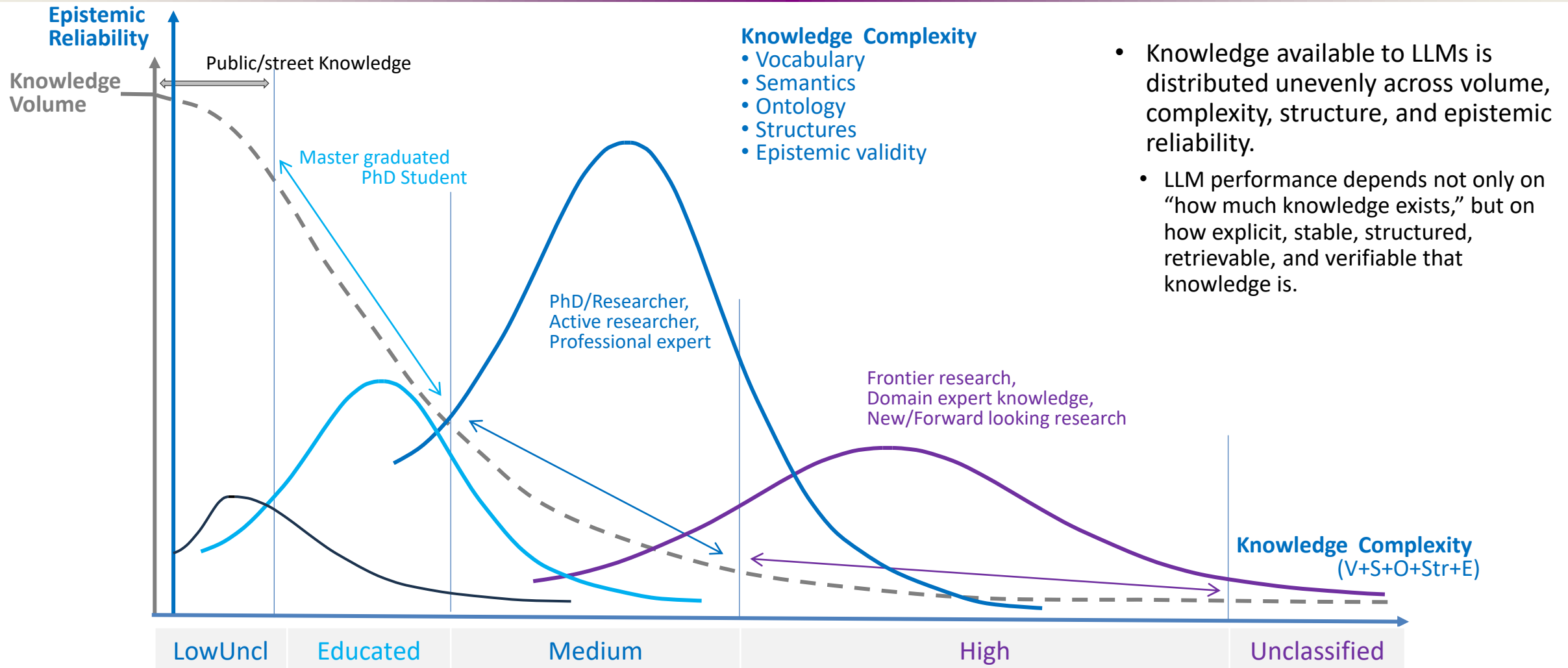


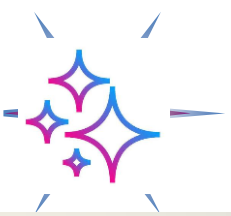
LLM accessible knowledge - Zones

- LLM-accessible knowledge can be modelled as a distribution across volume, structural complexity, and epistemic reliability.
- Zones:
 - **Public/common knowledge** occupies a high-volume but weakly validated region
 - **Educated knowledge** (university) and progressing professional and/or scientific (PhD student)
 - **Mature expert knowledge** occupies a smaller but more structured and reliable region
 - **Frontier research** occupies a highly complex but epistemically uncertain region.
 - (Not accessible to LLM) **Tacit knowledge** and professional experience (skills and aptitude + intuition)
- LLM performance is expected to be highest not at the extremes, but in domains where knowledge is *sufficiently abundant/rich, well-structured, and verifiable*.
- Effective researcher - LLM interaction should be studied as a function of knowledge complexity, epistemic-status complexity, and degree of human-guided verification.

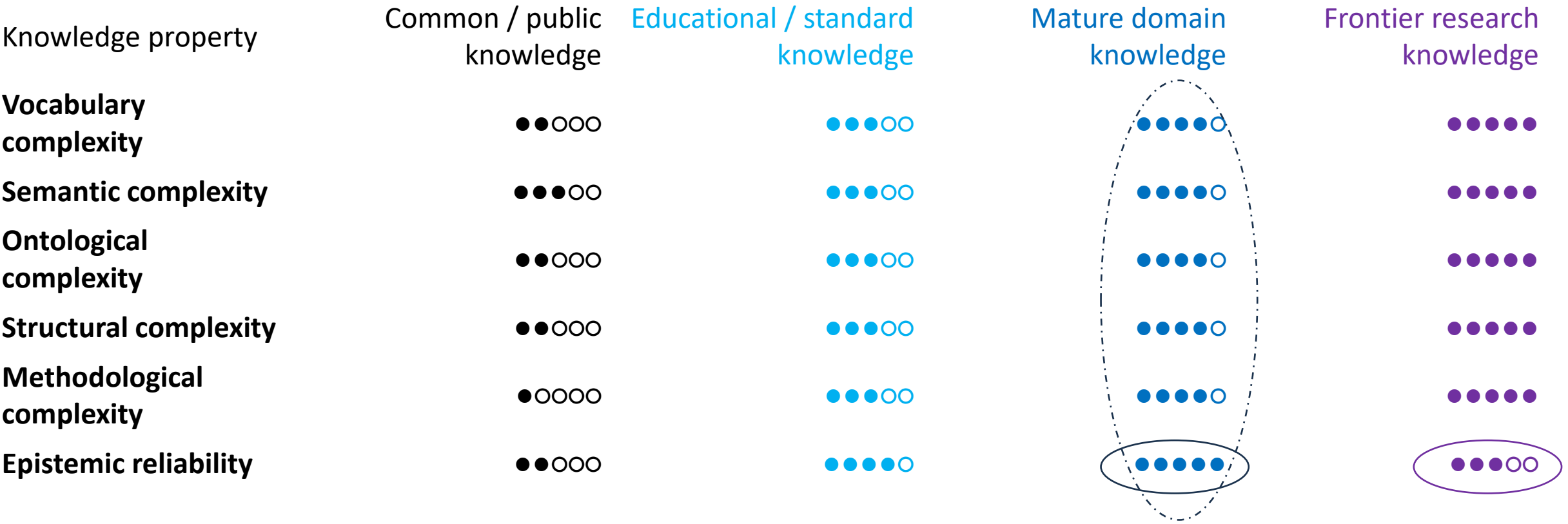


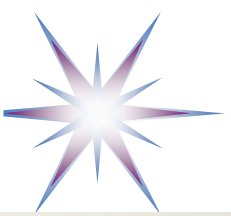
LLM Knowledge Curve(s) by Epistemic Reliability (1)





Tested with ChatGPT: Assessment Knowledge Complexity + Epistemic reliability by Knowledge Regions – Test @ beginning





Knowledge Complexity and Epistemic Governance Model of LLM-Assisted Research (KCEG)

Axis	Symbol	Description
Knowledge volume	V	Amount of available textual, data, and structured material
Knowledge complexity	C	Multidimensional complexity of vocabulary, semantics, ontology, structure, inference, method, and epistemic status
Epistemic reliability	R	Degree to which claims are validated, reproducible, consensual, and evidence-supported
Structural explicitness	S	Degree to which knowledge is organized in explicit forms: ontologies, taxonomies, graphs, methods, datasets
Frontier uncertainty	U	Degree of novelty, contestation, incompleteness, and instability
LLM support level	L	Capability level of the LLM system: base model, retrieval, tools, knowledge graph, verification
Researcher governance	G	Degree of expert involvement in framing, judging, verifying, and validating outputs
Effective answerability	A	Likelihood that the LLM system can generate useful, accurate, and epistemically appropriate outputs

A simplified model of effective LLM answerability

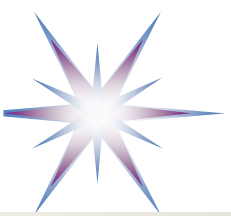
$$A = f(V, S, R, L, G) - g(C, U, \text{ambiguity, tacit knowledge})$$

Where:

- **A** = effective answerability or useful research support;
- **V** = volume of available knowledge;
- **S** = structural explicitness;
- **R** = epistemic reliability;
- **L** = LLM system capability;
- **G** = researcher governance;
- **C** = knowledge complexity;
- **U** = frontier uncertainty.

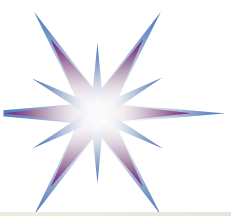
- Textual version:

- LLM usefulness increases when knowledge is sufficiently available, structured, reliable, retrievable, and governed by experts.
- It decreases when knowledge is highly ambiguous, novel, tacit, disputed, or weakly validated.



KCEG & HGLEP applied to and tested for practical research tasks

- Structured Systematic Literature Study (SSLR) – Knowledge complexity as metrics for paper assessment
- Software Complexity Model and Engineering Guidance (SCEG) and Agentic Development Environment
- Further/ongoing research:
 - Temporal aspect/dimension of KCEG, required knowledge body for education and training
 - Agentic Systems Interaction and Team Knowledge Management – as a challenge for software & applications developing companies now



Structured Systematic Literature Review (SSLR): Using KCEG Method for Paper Assessment (Template - detailed)

Paper information

- Paper title
- Publication place and date/year
- What is the paper about in one sentence?
- Main topic / problem addressed:
- Why is this paper relevant to my research?

Basic SLR

2. K1–K3: Vocabulary and factual inventory

- Key terms and phrases
- Main factual claims
- Where are these claims supported?

3. K3–K4: Concepts, entities, and roles

- Main concepts: What are the main concepts used by the authors?
- Definitions or implied meanings: Are the concepts clearly defined? Are any terms used vaguely?
- Entities, objects, or components: What kinds of things exist in the paper's research world?
- Roles and boundaries: What role does each main entity play?

Extended SLR

4. K4-K5: Structure and relations

- Relationship map
- Important relation types

Advanced SLR

5. K6-K8: Methodological reconstruction

- What method does the paper use?
- What data, tools, experiments, models, or analytical procedures are used?
- What is being measured, compared, tested, designed, or evaluated?

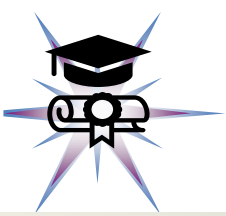
6. K7–K9: Inference, assumptions, limitations, and epistemic quality

- Main reasoning chain: How do the authors move from evidence to conclusion? Evidence → interpretation → claim → conclusion:
- Assumptions: What must be true for the paper's argument or method to work?
- Limitations: What are the main limitations, weaknesses, or boundary conditions?
- Uncertainty: What remains uncertain, under-evidenced, or open?

Advanced but partly used for Basic SLR (as presenting paper results)

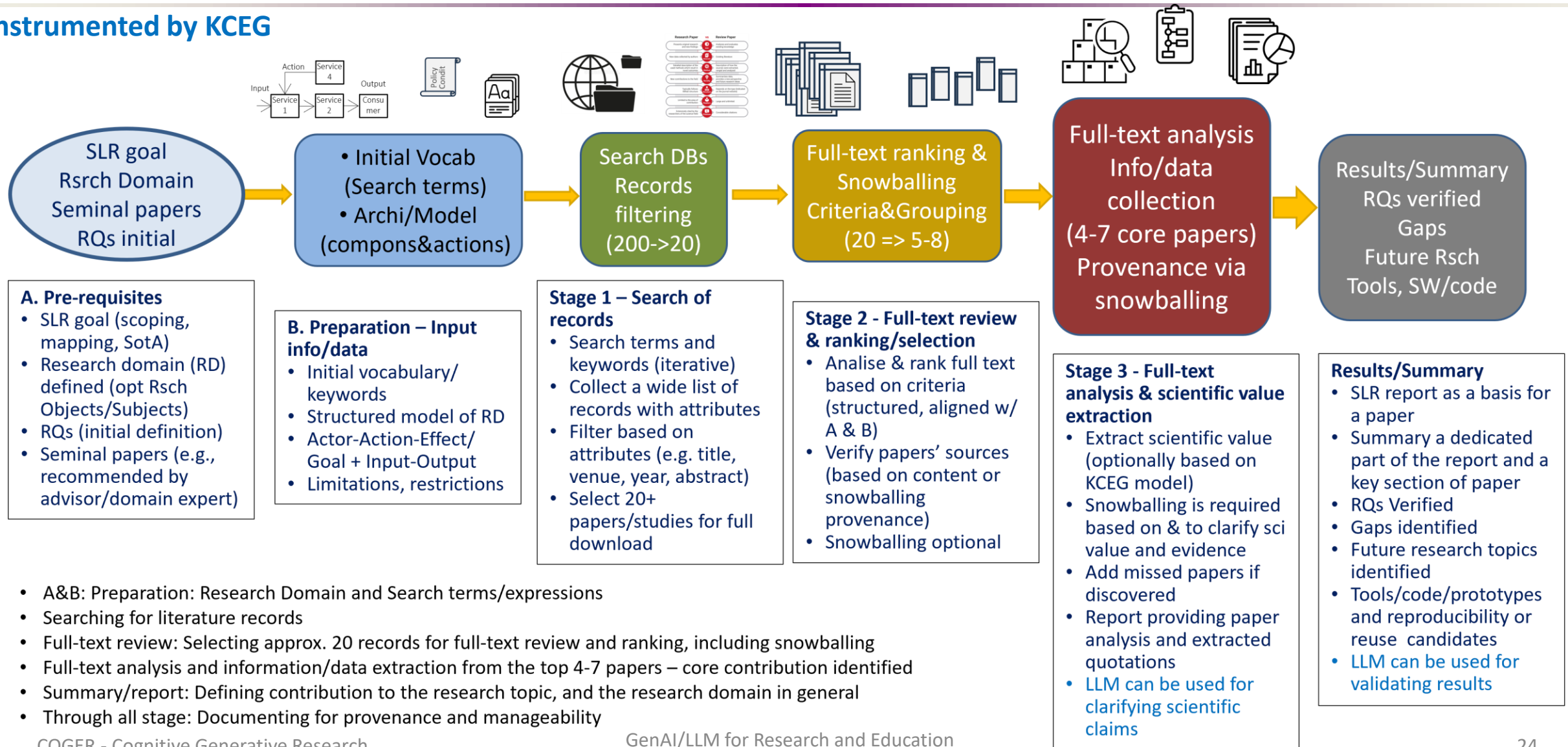
7. K9: Contribution and research trajectory

- What is the paper's main contribution?
- What is missing or underdeveloped?
- What research question could follow from this paper?
- How could this paper influence my own research trajectory?



Structured SLR for System and Software Engineering domain - Workflow as conducted by the PhD student & advised by the advisor (or domain expert)

Instrumented by KCEG

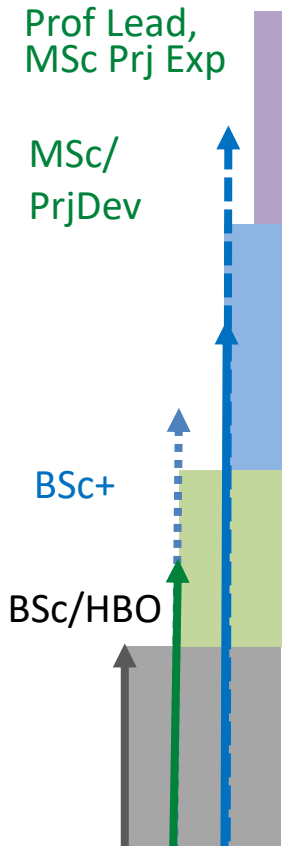




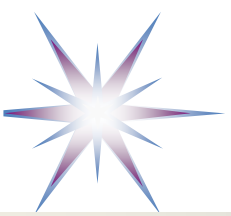
Applying KCEG Methodology to Software complexity Model (SCEG)

To provide a basis for System and Software Engineering Curriculum

code → modules → data models → APIs → architecture → quality attributes → runtime operation → lifecycle governance

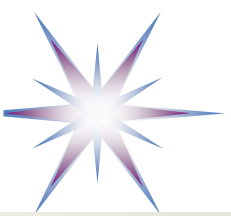


- SCEG-9 Evolution, governance, compliance, sustainability
- SCEG-8 Runtime, deployment, DevOps, observability
- SCEG-7 Quality attributes, verification, security, risk
- SCEG-6 Architecture styles, layers, views, decisions
- SCEG-5 APIs, contracts, data models, semantic interoperability
- SCEG-4 Modules, components, OO/functional design, patterns
- SCEG-3 Functional semantics, requirements traceability
- SCEG-2 Algorithms, data structures, computational logic
- SCEG-1 Code syntax, local implementation, style
- SCEG-0 Prerequisite: Intent, requirements, constraints, acceptance criteria



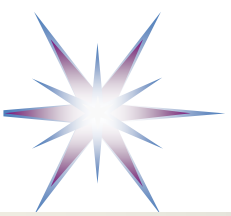
Applying KCEG and HGLEP to Agentic Systems Interaction and Team Knowledge Management – TODO In progress

- Linking and aligning existing frameworks and tools for Agents' interaction
 - MCP – Model Context Protocol (K1 – K4/KC1-KC2) (Anthropic, 25 November 2025)
 - OKF – Open Knowledge Format (Google, 12 June 2026)
- Introducing Agentic Knowledge Architecture (AKA)
 - What knowledge is available to the agentic system?
 - Knowledge supply, acquisition, storage, retrieval, context, lifecycle
 - Building Team/company knowledge corpus



Q&A

- Q: Can LLMs create knowledge?
- A: New framework allows a more mature answer:
 - “LLMs can help reorganize and **project accessible knowledge into candidate structures**, but knowledge creation becomes valid only through human-governed verification, methodological testing, and epistemic assessment.”
- A: Simple answer – LLM doesn’t create knowledge the same way as a researcher potentially gains knowledge horizon when interacting with LLM Assistant
 - A good research assistant should not answer confidently, but provide interactive support:
 - expose uncertainty
 - distinguish fact, inference, hypothesis, analogy, and speculation
 - help map concepts and claims
 - reveal gaps and contradictions
 - support verification
 - allow expert researchers to govern the process

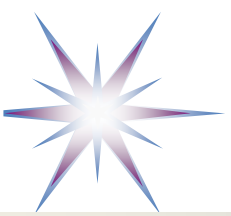


Scientific domains dealing with Knowledge

Traditional knowledge-dealing fields do define knowledge, but usually for their own operational purposes. They do not share one universal definition of knowledge, and they rarely provide a complete layered theory of “knowledge complexity” in the sense we are building for scientific artefact analysis.

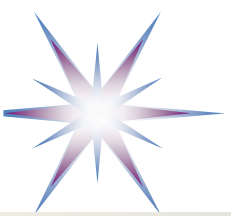
Different fields emphasise different aspects:

- **Knowledge Discovery / Data Mining (KDD):** knowledge as discovered valid, novel, useful, understandable patterns in data.
- **Knowledge Representation / AI (KP/AI):** knowledge as a formal symbolic or computational representation enabling reasoning.
- **Epistemology:** knowledge as justified, true, reliable, warranted belief.
- **Knowledge Management (KM):** knowledge as an organisational resource, practice, expertise, tacit/explicit know-how.
- **Knowledge Organisation / Library and Information Science:** knowledge as organised concepts, documents, classifications, and subject relations.
- **Cognitive science and learning sciences:** Cognitive science studies how humans represent, learn, reason with, and transform knowledge
- **Philosophy** and sociology of science: knowledge as theory-, method-, and community-validated scientific claim.



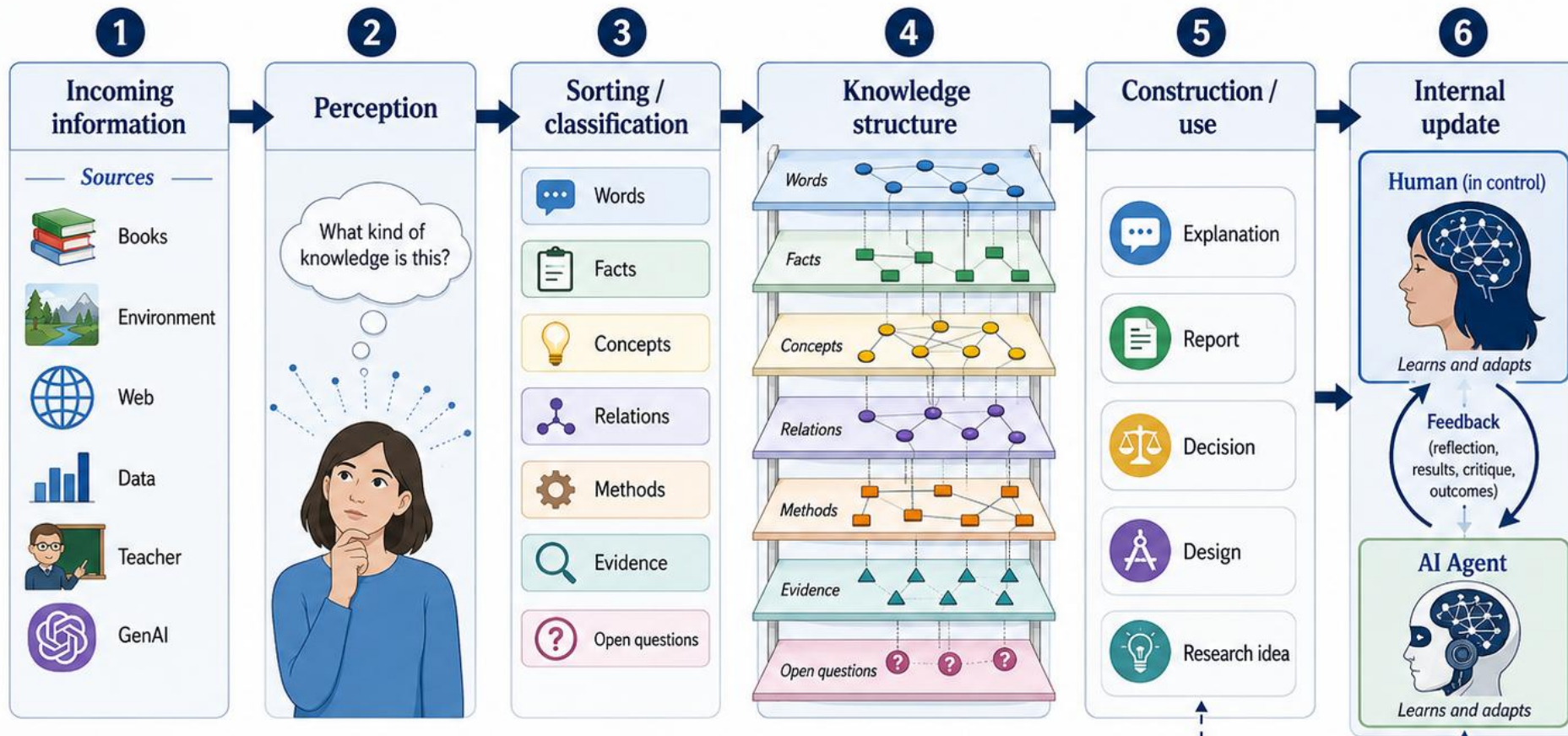
Epistemic Complexity – References (1)

- Epistemic complexity https://edutechwiki.unige.ch/en/Epistemic_complexity
- In various learning sciences, *epistemic complexity* often refers to the *cognitive challenge* of a task, in particular in constructivist whole task settings such as inquiry-based learning.
 - Lakkala, Minna (2010). How to design educational settings to promote collaborative inquiry: Pedagogical infrastructures for technology-enhanced progressive inquiry, Dissertation, Institute of Behavioural Sciences, University of Helsinki, Finland.
- Epistemic complexity indicates students' efforts to produce not only descriptions of the material world but also theoretical explanations and articulation of hidden mechanisms central to the nature of science
 - Salmon, W. C. 1984. Scientific explanations and the causal structure of the world. , Princeton, NJ: Princeton University Press
 - Zhang Jianwei; Marlene Scardamalia, Richard Reeve, Richard Messina (2009). Designs for Collective Cognitive Responsibility in Knowledge-Building Communities, *Journal of the Learning Sciences* Vol. 18, Iss. 1, 2009
- More simply: “Epistemic complexity represents the level of complexity at which a student chooses to approach an issue.” (Zhang, 2009)
- A philosopher defines epistemic complexity as the “richness of the knowledge that is embedded in an artifact.” (Dasgupta)
- In systems theory, “Rescher (1998) distinguished three 'modes', namely epistemic, ontological and functional complexity. Among these modes of complexity, the epistemic embraces three categories: descriptive, generative and computational complexity” (Schlindwein et al., 2004).
- A. Carsetti, Epistemic Complexity and Knowledge Construction: Morphogenesis, symbolic dynamics and beyond, Springer Nature, 2013, 150pp, ISBN: 978-94-007-6012-7
 - <https://link.springer.com/book/10.1007/978-94-007-6013-4>
- Hakkarainen, K. 2003. Progressive inquiry in a computer-supported biology class.. *Journal of Research in Science Teaching* , 40: 1072–1088. [DOI: 10.1002/tea.10121](https://doi.org/10.1002/tea.10121)



Extra Illustration Graphics

KCEG as a Lens for Human-Agent Interaction



perception
Taking in information through the senses and attention.



representation
Classifying and labeling information into meaningful categories.



presentation
Organizing knowledge in a structured, connected form.



knowledge projection
Using structured knowledge to produce outputs and drive action.

Using KCEG (Knowledge Complexity & Effective Guidance) Model

* For structured Perception, Learning and building Knowledge corpus



The human remains responsible for structure, judgment, and update.



Using KCEG (Knowledge Complexity & Effective Guidance) Model

* For structured Perception, Learning and building Knowledge corpus



Sort information while it arrives – this is how your mind builds knowledge.

